

LAPORAN PRAKTIKUM

WUOKSHOP SISTEM INFORMASI

WEB SERVER

(TIF330805)

SEMESTER 3



Acara 7

(Database, CREATE TABLE, dan Model Dasar)

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TEKNIK INFORMATIKA KAB. NGANJUK

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POLITEKNIK NEGERI JEMBER

TAHUN 2026

- **Studi Kasus**

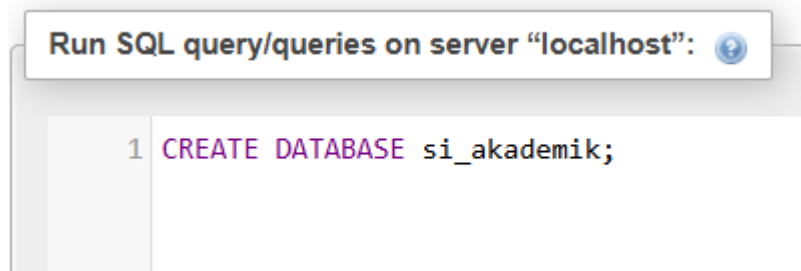
Sistem Informasi Akademik membutuhkan data Mahasiswa, Program Studi, dan Mata Kuliah. Data ini harus tersimpan permanen, bisa dicari, diubah, dan dihapus. File teks atau array PHP tidak cocok untuk data yang besar dan sering berubah. Database relasional (MySQL) adalah solusinya.

- **Tujuan**

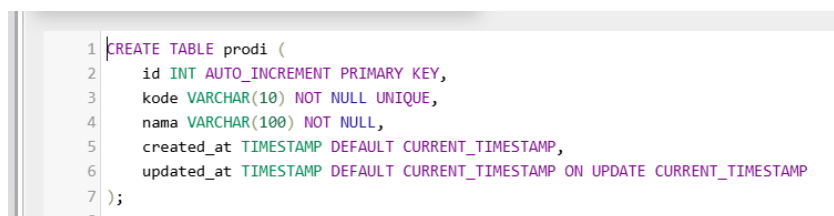
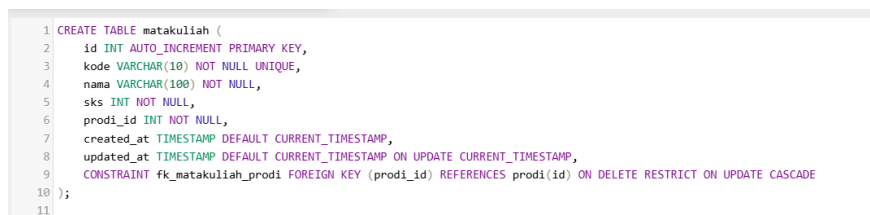
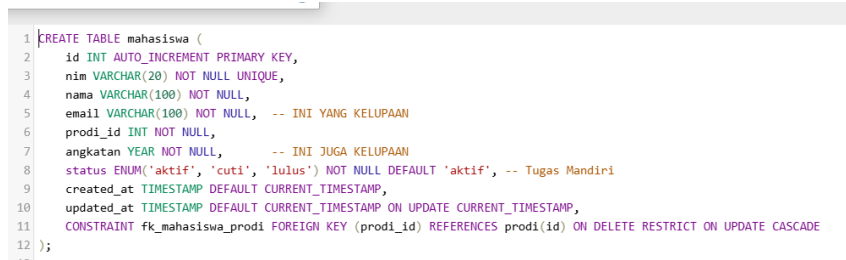
Mahasiswa membuat database si_akademik, tiga tabel, data awal, dan Model sederhana yang bisa mengambil semua data.

- **Langkah**

1. Buka phpMyAdmin, buat database si_akademik.



2. Jalankan semua perintah CREATE TABLE.



3. Jalankan INSERT data awal.

```
Run SQL query/queries on database si_akademik: ⓘ

1 INSERT INTO mahasiswa (nim, nama, email, prodi_id, angkatan, status) VALUES
2 ('240801', 'Budi Santoso', 'budisantoso@gmail.com', 1, 2024, 'aktif'),
3 ('240802', 'Ani Wijaya', 'aniwijaya@gmail.com', 1, 2024, 'aktif'),
4 ('240803', 'Citra Lestari', 'citralestari@gmail.com', 2, 2024, 'aktif');
5
```

```
Run SQL query/queries on database si_akademik: ⓘ

1 INSERT INTO matakuliah (kode, nama, sks, prodi_id) VALUES
2 ('TI101', 'Pemrograman Dasar', 3, 1),
3 ('TI102', 'Basis Data', 3, 1),
4 ('SI101', 'Pengantar SI', 2, 2);
```

```
Run SQL query/queries on database si_akademik. ⓘ

1 INSERT INTO prodi (kode, nama) VALUES
2 ('TI', 'Teknik Informatika'),
3 ('SI', 'Sistem Informasi'),
4 ('TK', 'Teknik Komputer');
5
```

4. Buat config/database.php.

```
folder to enable all features. Manage Learn More

detail.php  Mahasiswa.php  Dosen.php

si-akademik7 > config > database.php

1 <?php
2 return [
3     'host' => 'localhost',
4     'dbname' => 'si_akademik',
5     'username' => 'root',
6     'password' => '',
7     'charset' => 'utf8mb4',
8 ];
9 ?>
```

5. Buat class Model dasar dan MahasiswaModel yang punya method all().

```

si-akademik7 > app > core > Model.php > Model
1  <?php
2  namespace App\Core;
3
4  use PDO;
5  use PDOException;
6
7  class Model {
8      protected $db;
9
10     public function __construct() {
11
12         $config = require __DIR__ . '/../../config/database.php';
13
14         $dsn = "mysql:host={$config['host']};dbname={$config['dbname']};charset={$config['charset']}";
15
16         try {
17             $this->db = new PDO($dsn, $config['username'], $config['password']);
18             $this->db->setAttribute(PDO::ATTR_ERRMODE, PDO::ERRMODE_EXCEPTION);
19             $this->db->setAttribute(PDO::ATTR_DEFAULT_FETCH_MODE, PDO::FETCH_ASSOC);
20         } catch (PDOException $e) {
21             die("Koneksi database gagal: " . $e->getMessage());
22         }
23     }
24 }

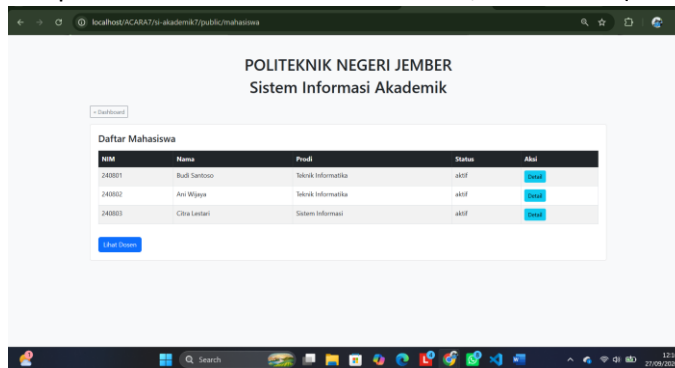
```

```

si-akademik7 > app > models > Mahasiswa.php > Mahasiswa
1  <?php
2  namespace App\Models;
3
4  use App\Core\Model;
5  use PDO;
6
7  class Mahasiswa extends Model {
8
9      public static function all() {
10         $instance = new self();
11         $stmt = $instance->db->query("
12             SELECT m.nim, m.nama, m.status, p.nama AS prodi
13             FROM mahasiswa m
14             LEFT JOIN prodi p ON m.prodi_id = p.id
15         ");
16         return $stmt->fetchAll();
17     }
18
19     public static function findByNim($nim) {
20         $instance = new self();
21         $stmt = $instance->db->prepare("
22             SELECT m.nim, m.nama, m.status, p.nama AS prodi
23             FROM mahasiswa m
24             LEFT JOIN prodi p ON m.prodi_id = p.id
25             WHERE m.nim = :nim
26         ");
27         $stmt->execute(['nim' => $nim]);
28         return $stmt->fetch();
29     }
30 }

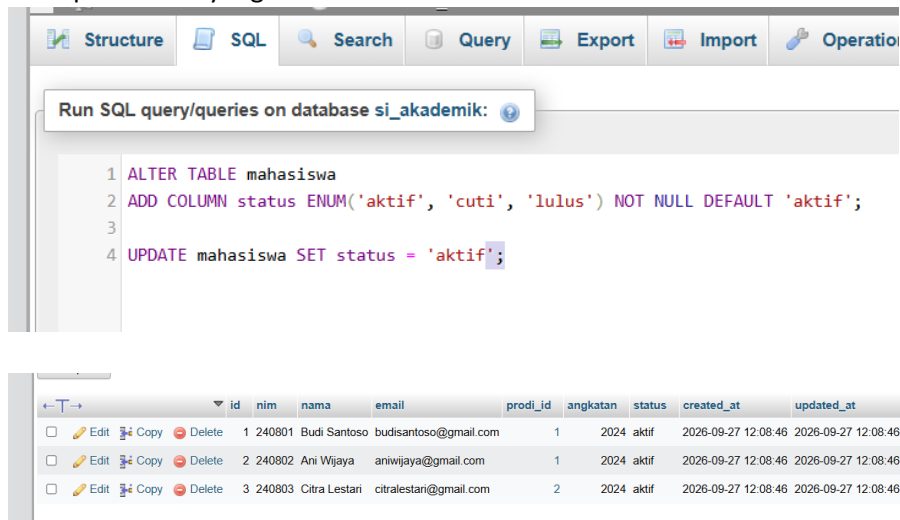
```

6. Tampilkan data mahasiswa di halaman /mahasiswa (masih sederhana).



- Tugas Mandiri

Tambahkan kolom 'status' (ENUM 'aktif','cuti','lulus') pada tabel mahasiswa, isi default 'aktif', lalu update data yang sudah ada.



The screenshot shows a database management interface with a menu bar (Structure, SQL, Search, Query, Export, Import, Operation) and a toolbar. A panel titled "Run SQL query/queries on database si_akademik:" contains the following SQL queries:

```
1 ALTER TABLE mahasiswa
2 ADD COLUMN status ENUM('aktif', 'cuti', 'lulus') NOT NULL DEFAULT 'aktif';
3
4 UPDATE mahasiswa SET status = 'aktif';
```

Below the queries, a table view displays the data for the 'mahasiswa' table. The table has columns: id, nim, nama, email, prodi_id, angkatan, status, created_at, and updated_at. There are three rows of data, all with 'status' set to 'aktif'.

	id	nim	nama	email	prodi_id	angkatan	status	created_at	updated_at
☐ Edit Copy Delete	1	240801	Budi Santoso	budisantoso@gmail.com	1	2024	aktif	2026-09-27 12:08:46	2026-09-27 12:08:46
☐ Edit Copy Delete	2	240802	Ani Wijaya	anijijaya@gmail.com	1	2024	aktif	2026-09-27 12:08:46	2026-09-27 12:08:46
☐ Edit Copy Delete	3	240803	Citra Lestari	citralestari@gmail.com	2	2024	aktif	2026-09-27 12:08:46	2026-09-27 12:08:46